Harder Geometrical Problems — problem 12.3.2.50

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Prove that if the opposite sides of a skew quadrilateral are congruent, then the line joining the midpoints of the two diagonals is perpendicular to these diagonals, and conversely, if the line joining the midpoints of the two diagonals of a skew quadrilateral is perpendicular to these diagonals, then the opposite sides are congruent. (This is again USO 1977. Now we are looking for a short geometric solution.)